

MD. SHAON KHAN

AI/ML & Software Engineering Intern Candidate

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SUMMARY

3rd-year Information Technology undergraduate at the Institute of Information Technology (IIT), Jahangirnagar University, with 2+ years of hands-on experience in AI/ML, IoT, and full-stack software development. Developed machine learning applications for healthcare and agriculture, integrated IoT devices with cloud-based software systems, and built end-to-end web applications using modern backend technologies. Seeking an AI/ML or Software Engineering internship to contribute technical skills in real-world product development.

EDUCATION

Institute of Information Technology (IIT), Jahangirnagar University

Savar, Dhaka

B.Sc. in Information Technology — 3rd Year, 1st Semester 2022 – Present

- CGPA: 3.82 / 4.00 (up to 4th Semester)

Jahangirnagar University School and College

Savar, Dhaka

- Higher Secondary Certificate (HSC) – Science | GPA: 5.00 / 5.00 | 2022
- Secondary School Certificate (SSC) – Science | GPA: 5.00 / 5.00 | 2020

TECHNICAL SKILLS

- **Languages:** Python, C, C++, Java, JavaScript
- **Machine Learning:** TensorFlow, PyTorch, Keras, Scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Seaborn
- **GenAI / LLM:** LangChain, Retrieval-Augmented Generation (RAG)
- **Backend / Frontend:** FastAPI, Node.js, HTML5, CSS3, Bootstrap
- **Database:** MySQL, PostgreSQL, MongoDB Atlas
- **IoT & Signal Processing:** Arduino Uno, ESP8266, AD8232 ECG Sensor, Bandpass/Notch Filtering, Wavelet Denoising, Baseline Wander Removal
- **Deployment & Tools:** Render, Netlify, Vercel, Git, GitHub, VS Code, Jupyter Notebook, Windows/Linux

MACHINE LEARNING & SOFTWARE PROJECTS

HealthBridge — IoT Health Monitoring & Disease Prediction

GitHub: github.com/Md-Shaon-Khan/HealthBridge-with-ECG-Check-Up

- Built a full-stack IoT healthcare platform integrating ECG acquisition with deep learning-based cardiac disease prediction.
- Trained ResNet1D, InceptionTime, and DenseNet121 models on the MIT-BIH, PTB-XL, and CPSC 2018 datasets for eight cardiac condition classes.
- Implemented a multi-stage ECG signal preprocessing pipeline: DC offset removal, baseline wander removal, notch filtering, bandpass filtering, and wavelet denoising.
- Developed a patient-doctor dashboard and integrated a medical chatbot; deployed the FastAPI backend on Render and the frontend on Netlify.

Farmer's Friend (কৃষক বন্ধু) — Paddy Disease Classifier

GitHub: github.com/Md-Shaon-Khan/Farmer-s-Friend

- Developed a deep learning ensemble using EfficientNetB0, ResNet50, and DenseNet121 trained on ~49,000 images collected from six datasets.
- Implemented a binary gatekeeper model to reject non-paddy image inputs before disease classification.
- Achieved ~97% validation accuracy across five rice leaf disease classes.
- Served as the foundation for ArogyoAI, a Top 100 Finalist project at Infinity AI BuildFest 2026.

IIT Academic Website — Batch-wise Academic Portal

GitHub: github.com/Md-Shaon-Khan/IIT-Academic-Website-with-Backend

- Developed a batch-wise academic portal for the Institute of Information Technology using FastAPI and MySQL.
- Enabled administrators to manage technical projects, GitHub repositories, supervisors, and image galleries.
- Implemented search functionality for academic projects and designed a responsive interface for examination papers, project archives, and academic calendars.

UniFix — University Complaint Management System

GitHub: github.com/Md-Shaon-Khan/UniFix

- Developed a full-stack complaint management platform with real-time status updates using Socket.io.
- Implemented role-based authentication, issue categorization, and secure database management using MySQL.
- Built dashboards for students and administrators to streamline campus issue reporting and resolution.

TECHNICAL EXPERIENCE

- **AI/ML Development (1+ Year):** End-to-end ML/DL workflows (EDA, preprocessing, feature engineering, hyperparameter tuning) using Scikit-learn, TensorFlow, and Keras across real-world datasets.
- **IoT & Signal Processing:** Integrated Arduino Uno, ESP8266, and AD8232 ECG sensors with AI-powered backend systems, including full ECG signal preprocessing.
- **Software Development:** Built and deployed full-stack web apps (FastAPI, Node.js, MySQL, PostgreSQL) with RESTful APIs, role-based auth, and real-time features via Socket.io.

Hackathon Experience

- Top 100 Finalist, Infinity AI BuildFest 2026, BRAC University.
- Contributed as an AI & IoT Developer in developing ArogyoAI, an AI-assisted healthcare solution for predictive diagnostics and decision support.

Competitive Programming

- Solved 400+ algorithmic problems across Codeforces (300+) and LeetCode (100+) using C++ and Python.
- Achieved a peak Codeforces rating of 1182 while strengthening problem-solving, data structures, and algorithm design skills.

Professional Training — Phitron, AI/ML Engineering

- Completed project-based training covering Python programming, statistics, EDA, supervised learning, unsupervised learning, deep learning, CNNs, and modern NLP concepts.
- Implemented hands-on projects covering regression, classification, clustering, computer vision, and end-to-end machine learning workflows using real-world datasets.

CERTIFICATIONS

- Kaggle Learn: Intro to Machine Learning
- Kaggle Learn: Intermediate Machine Learning
- SimpliLearn: PostgreSQL Professional Certification
- HackerRank: Python (Basic)